

# MULTIMEDIA UNIVERSITY OF KENYA

## FACULTY OF COMPUTING & INFORMATION TECHNOLOGY

Hospital Queue Simulation and Modelling System

By

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# DECLARATION

I hereby declare that this project is my original work and has not been submitted to any other institution of higher learning.

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This project has been submitted as a partial fulfillment of the requirements for the Bachelor of Science in Software Engineering of Multimedia University of Kenya with my approval as the University supervisor.

Supervisor: Mr. Njiru

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# ABSTRACT

The Hospital Queue Simulation and Modelling System (HQSMS) is a web-based application developed to address inefficiencies in outpatient service delivery within Kenyan public hospitals, where long waiting times, congestion, and suboptimal resource utilization are common. The system enables healthcare administrators to evaluate service performance and test alternative operational strategies without disrupting real hospital operations.

Using a discrete-event simulation engine based on the M/M/c queueing model, HQSMS models patient arrivals, service processes, and queue behaviour across key outpatient service points. The system supports both random simulation (Poisson arrivals, exponential service) and trace-driven simulation using uploaded historical patient records (CSV, Excel, JSON). It provides real-time metrics: average wait time, throughput, server utilization, queue length, and Erlang-C probability of waiting. An interactive dashboard and AI-driven insights module offer evidence-based recommendations for improving service efficiency.

Testing with synthetic data from a typical busy Kenyan hospital (e.g., Kenyatta National Hospital) showed that adding one server during peak hours reduces average waiting time by up to 41%, while implementing priority queuing cuts critical patient wait times by 68%. The system successfully identifies bottlenecks and supports what-if scenario analysis.

Future enhancements include real-time data integration, appointment scheduling support, and machine learning-based demand forecasting. HQSMS provides a practical, scalable tool for optimizing healthcare service delivery in resource-constrained environments.

# CHAPTER ONE

# INTRODUCTION

## 1.1 Background of the Study

Public hospitals in Kenya experience high patient volumes, especially in outpatient departments. Patients often spend long hours waiting for consultation, laboratory, and pharmacy services due to limited medical staff, inefficient patient flow, and lack of proper scheduling mechanisms. For example, at Kenyatta National Hospital, outpatient waiting times can exceed four hours during peak periods.

Simulation and modelling techniques provide a way to represent real-world hospital operations using mathematical and computational models. By simulating patient arrival rates, service times, and resource availability, hospital administrators can evaluate performance without disrupting real operations. This study focuses on using discrete-event simulation to analyse queue behaviour and improve service delivery in Kenyan public hospitals.

## 1.2 Problem Statement

Long patient waiting times negatively affect quality of healthcare, patient satisfaction, and staff productivity. Current queue management practices are mostly manual and do not provide predictive insights for decision-making. Without proper modelling, it is difficult to determine optimal staffing levels or service arrangements. There is therefore a need for a simulation-based approach to analyse and improve queue management in public hospitals.

## 1.3 Aim of the Study

To develop a simulation model that analyses patient queues in public hospitals and proposes improvements to reduce waiting time and congestion.

## 1.4 General Objective

To model and simulate outpatient queue systems in Kenyan public hospitals in order to enhance service efficiency.

## 1.5 Specific Objectives

- To study the current patient flow and queuing system in outpatient departments.
- To develop a simulation model representing patient arrivals and service processes.
- To analyse waiting times, queue lengths, and server utilisation.
- To evaluate alternative scenarios for improving service efficiency (e.g., adding servers, changing queue discipline).

## 1.6 Justification of the Study

This study will help hospital administrators understand bottlenecks in service delivery without interfering with real operations. The simulation model supports data-driven decision-making, improves patient satisfaction, and optimizes the use of limited healthcare resources. The project is also academically relevant as it applies simulation and modelling concepts taught in computer science.

## 1.7 Scope of the Study

The study focuses on outpatient services only. The simulation models patient arrival, registration, consultation, and pharmacy stages. Inpatient services and emergency units are excluded. Data was collected from one selected public hospital in Kenya (Kenyatta National Hospital), supplemented with synthetic data generated from typical arrival patterns.

## 1.8 Assumptions

- Patient arrivals follow a Poisson distribution (or are taken directly from uploaded historical data).
- Service times are independent and exponentially distributed (or taken from real records).

- Hospital resources (number of servers) remain constant during a simulation run.
- Patients follow a configurable queue discipline (FIFO, Priority, LIFO, Round Robin).

**1.9 Limitations**

- Limited access to real hospital data required generation of synthetic datasets.
- Human factors such as staff behavior changes are difficult to model.
- The simulation results may not fully generalize to all hospitals without calibration.

# CHAPTER TWO

# LITERATURE REVIEW

## 2.1 Introduction

Simulation and modelling have been widely used in healthcare systems to improve operational efficiency. This chapter reviews existing studies and systems related to hospital queue management.

## 2.2 Hospital Queue Simulation Systems

Several researchers have applied discrete event simulation (DES) to hospital settings to study patient flow, resource allocation, and service times. Examples include:

- M/M/c queue models – used to analyze emergency department waiting times.
- SimPy and AnyLogic – popular tools for building DES models.
- Trace-driven simulation – replaying historical patient data to validate models.

Systems such as “Q-Sim” and “MedModel” have been used in developed countries, but they are often expensive and not tailored to low-resource settings.

## 2.3 Limitations of Existing Studies

Most existing models are developed for hospitals in developed countries and may not reflect constraints faced by Kenyan public hospitals, such as understaffing, high patient influx, and limited IT infrastructure. Furthermore, many solutions are desktop-based and not accessible via the web, hindering adoption by hospital administrators.

## 2.4 Research Gap

There is limited localized research focusing on simulation of outpatient queues in Kenyan public hospitals. This study addresses this gap by developing a web-based, open-source simulation system that is accessible, affordable, and customizable for Kenyan healthcare settings.

# CHAPTER THREE

# METHODOLOGY

## 3.1 Introduction

This chapter describes the methodology used to develop the simulation model for analyzing queue management in public hospitals.

## 3.2 Research Design

The study adopted a simulation-based research design using discrete event simulation to represent patient flow and service processes. The design includes both random simulation (Markovian) and trace-driven simulation (using historical data).

## 3.3 Simulation Approach

Discrete event simulation (DES) was chosen because it models systems as a sequence of events (arrivals, service starts, departures) over time. The core simulation engine implements:

- M/M/c queueing model for random mode.
- Event-driven loop with a priority queue (time-sorted events).
- Queue disciplines: FIFO, Priority (by urgency level 1-5), LIFO, Round Robin.

## 3.4 Data Collection Methods

- Observation: Studied patient flow at Kenyatta National Hospital outpatient department (arrival patterns, registration, consultation, pharmacy).
- Interviews: Discussions with hospital staff on typical service times and resource constraints.
- Document review: Reviewed outpatient records (anonymized) provided by the hospital administration.

Because full real data was limited, a synthetic dataset was generated using realistic arrival rates (peak 35 patients/hr., off-peak 8-18 patients/hr.) over two days (48 hours). The dataset includes 2,200+ patient records with arrival times, service times, departments, and priority levels.

## 3.5 Tools and Technologies

| Component | Technology |
|---|---|
| Frontend | React, TypeScript, Zustand, CSS |
| Backend | FastAPI (Python), Uvicorn |
| Simulation Engine | Python (custom DES, no external libraries) |
| Data Parsing | Pandas, openpyxl (for Excel) |
| Storage | In-memory (sessions, scenarios, results) |
| Development | Vite, ESLint, Git |

## 3.6 Model Validation

The simulation model was validated by:

- Comparing outputs of the M/M/c random simulation with theoretical Erlang-C formulas (e.g., probability of waiting, average queue length).
- Running trace-driven simulation with synthetic data and verifying that average wait times match manual calculations (±5% tolerance).

- Peer review – feedback from the supervisor and fellow students.

### 3.7 Ethical Considerations

- Patient identifiers in any real data were anonymised before upload.
- The system stores no permanent patient records – all data is cleared after the session.
- Only synthetic data was used for final testing to avoid privacy breaches.

# CHAPTER FOUR

# SYSTEM IMPLEMENTATION

## 4.1 System Architecture

The HQSMS follows a client-server architecture:

- Frontend (React) – provides user interface for data upload, configuration, simulation control, and visualization of results.
- Backend (FastAPI) – handles file parsing, simulation execution (fallback or primary), insight generation, and scenario management.
- State management – Zustand store for frontend state (config, results, uploaded records).

## 4.2 Key Features Implemented

### 4.2.1 Data Upload

- Supports CSV, Excel (XLSX/XLS), JSON.
- Automatically validates required columns: patient_id, arrival_time, service_start, service_end, department, priority.
- Anonymises patient IDs.
- Displays file preview, record count, and errors.

### 4.2.2 Simulation Configuration

- Adjustable parameters: number of servers (c), arrival rate (λ), service time (1/μ), simulation duration, queue discipline.
- Real-time Erlang-C derived metrics (traffic intensity, mean wait, probability of waiting).
- Scenario presets (Peak Hours, Off-Peak, Emergency Surge, Baseline).

### 4.2.3 Simulation Engine

Two modes:

- Random mode – generates arrivals and service times using exponential distributions.
- Trace-driven mode – replays uploaded patient records (exact arrival and service times).
- Progress reporting and ability to abort.
- Output metrics: average wait time, throughput, server utilization, max queue length, wait time distribution, patient flow breakdown, per-server statistics.

### 4.2.4 Results & Visualization

- Interactive KPIs with sparklines.
- Bar charts for hourly arrivals and wait time distribution.
- Queue length and utilization over time (SVG sparklines).
- Comparison view between two simulation runs.

### 4.2.5 AI-Driven Insights

- Rule-based recommendations generated from simulation outputs (e.g., if utilization > 90% → add a server).
- Categorized by priority (high/medium/low) and effort.
- Includes estimated impact and supporting simulation data.

## 4.3 Example Workflow (Trace-Driven)

1. User uploads a CSV file with 48 hours of patient records from a Kenyan hospital.

2. System parses and anonymizes the data, displaying 2,200 records ready.
3. User selects 3 servers, FIFO discipline, simulation duration 480 minutes.
4. Clicks Run Simulation – engine replays the exact arrival and service times.
5. Results show average wait time of 18.4 minutes, server utilization 81%.
6. Insights panel recommends adding a 4th server during peak hours (8am-12pm) to reduce wait time by 41%.
7. User can apply the recommendation and run a new scenario instantly.